Remote photoplethysmography for heart rate and blood oxygenation measurement: a Review

Pireh Pirzada, Adriana Wilde, and David Harris-Birtill

***Abstract*—Usually measured through obtrusive contact-based methods, heart rate (HR) and peripheral oxygen saturation (SpO_2) are critical physiological signs used by clinicians during emergency interventions. Remote Photoplethysmography (rPPG) allows for unobtrusive sensing of these vital signs for health monitoring in various settings. We present a review of rPPG-related research conducted including related processes and techniques, such as regions of interest (ROI) selection, extracting the raw signal, pre-processing data, applying noise reduction algorithms, Fast Fourier transforms (FFT), filtering and extracting these vital signs. Further, we present a detailed, critical evaluation of available rPPG systems. Limitations and future directions have also been identified to further advance this field.**

***Index Terms*—Remote Photoplethysmography (rPPG), Heart Rate, Blood Oxygenation, Signal Processing, Health Monitoring.**

I. Introduction

HEART Rate (HR) and blood oxygen saturation (SpO_2) are physiological signs used as critical indicators of human health, as they are clinical alerts of a significant change [1]. The first of these signs, HR, is a critical measure of cardiac activity [2], used for assessment of emergent ill-health. At rest, normal values lie between 60-100 Beats Per Minute (BPM) [3]. Lower values ('bradycardia') might indicate hypovolemic shock [4], and higher ones ('tachycardia') [3], high intracranial pressure [5].

The second sign, SpO_2, is a broad measure of blood oxygen saturation, the percentage of Haemoglobin (Hb) in the blood carrying oxygen. When a person breathes, oxygen enters the lungs, transfers across the alveolar membrane and into the bloodstream [6], allowing oxygen transfer to all body cells [7]. Once the oxygen has been off-loaded, the Hb becomes deoxygenated haemoglobin, 'deoxy-Hb' [7], returning to the heart which then sends it on to the lungs to acquire more oxygen, and so the process continues [2]. Abnormal changes in SpO_2 can affect organs and is often associated with disease [7], such as pneumonia [8], asthma [9], chronic obstructive pulmonary disease (COPD) [10], [11], hypoxia [12], drug overdose [12], respiratory [13] or heart diseases [14]. SpO_2 values below 94% are abnormal for most people, although some individuals with stable chronic conditions may have readings of 88-94% [15]. Monitoring changes in SpO_2 can help identify complications including respiratory failure [13]. Here, we introduce the measurement of HR and SpO_2, explaining the methods used for these measurements, including contact-based pulse oximetry, and their limitations, which motivate the use of remote photoplethysmography (rPPG), as an innovative approach to overcome them.

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A. Measuring HR and SpO_2

Measurement of these vitals impacts clinical decisions [16]. For HR, electrocardiograms (ECG) are used to characterise the cardiac rhythm in detail but require keeping electrodes in place, typically with gel patches or chest straps [17], often requiring shaving body hair and adherent tape, which may irritate the skin or cause allergic reactions [18], [19] especially in long-term monitoring [19]. For SpO_2, invasive procedures like Arterial Blood Gas (ABG) sampling can be used. Here, blood is drawn from an artery [20], which is painful and adds a risk of infection [15], [20]. Regular sampling is time-consuming [21] and adds a risk of blood loss in neonatal care [22]. These vitals can also be monitored with contact-based pulse oximetry [23], [24], [25], discussed next.

B. Contact-based pulse oximetry

Pulse oximetry is based on the photoplethysmography (PPG) principles introduced by Hertzman in 1937 [26]. Blood flow is measured through changes in light dispersion [27], [28]. Pulse oximetry has a Red and Infrared (IR) light-emitting diode (LED) placed on one side of the probe and a photodetector on the other side. Pulse oximetry uses two wavelengths of light through the finger to a photodetector where light is absorbed by the finger and the rest reaches the photodetector. The light absorption is governed by Beer's law [29], i.e., proportionately to the concentration of the light-absorbing material (e.g. Hb and deoxy-Hb), hence, the higher their concentration, the more light is absorbed. As light-absorbing materials, Hb and deoxy-Hb differ: Hb absorbs more IR light whereas deoxy-Hb absorbs more red light as shown in Figure 1 [30]. Pulse oximeter LEDs emit light at 660 nm (red) and 940 nm (IR) wavelengths [31].

Pulse oximeters are the cheapest and most common way to measure HR and SpO_2 [33]. They are comparatively less intrusive in clinical settings [25], as they can be clipped on a fingertip, an earlobe [34], wrapped on a baby's foot, on a wrist band [35], or even stuck to the forehead [36]. Portable pulse oximeters have now become adopted worldwide for self-evaluating health vitals at home [37] and played an important role during the COVID-19 pandemic [38]. They are also used for sports [39], [40], and devices may have Bluetooth connectivity with smartphones, allowing data recording over

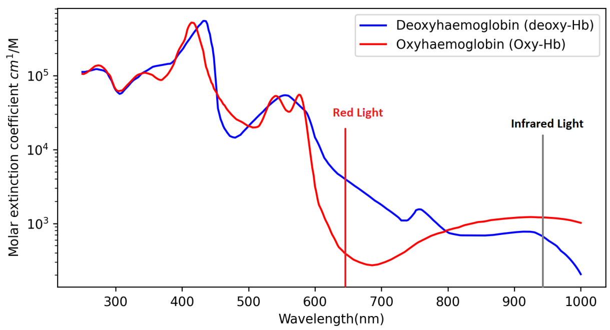

Fig. 1. The optical extinction spectra of oxy-Hb and deoxy-Hb within the blood. Figure generated using data by Prahl [32], [30].

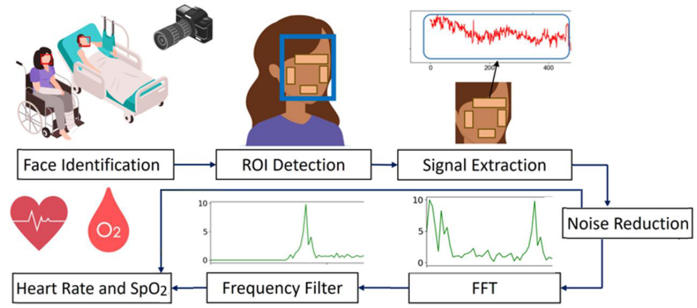

Fig. 2. rPPG process flow shown in the image from identifying the faces of people to obtaining their heart rate and blood oxygenation levels remotely (Vector images by Freekpik and graph images generated by the authors.)

time [41], [42], [43]. Commercial wearable devices (e.g., Apple Watch 6) use PPG [44], but their accuracy decreases in the active state compared to resting state [45]. These solutions require each patient to have such a device which can be costly, burdensome and time-consuming to set up for each patient for example in a triage setting for clinicians to monitor multiple patients simultaneously. Additionally, contact-based devices encounter challenges, as mentioned below, when used with patients who have physical or cognitive impairments.

Although effective, pulse oximetry presents some accuracy problems as a monitoring tool [20], [15]. A pulse oximeter must be physically placed on a finger, which is hard for some patients with incapacities or cognitive challenges [46], [47], [48], during surgery [47], sensitive skin [18], burnt skin [47], and in those with severe illness [18] as well as infants [49], [50] and young children [22]. It may be cumbersome to wear for long periods of time [51], restricting movement if wire-based pulse oximeter devices are used [36] and providing a constant physical reminder that their health is being monitored [18], [52]. It also increases risk of cross-infection [53].

Accuracy of pulse oximeters has been reported to be low for SpO_2 values under 83% [7] and is affected by nail varnish, skin tone, access movement, intravascular dyes and having abnormal levels of Hb and Carboxyhaemoglobin [7], [46]. Research shows that for participants with dark skin the error rate is higher [54], [55]. One study found a substantial bias in pulse oximeter devices used with people with darker skin tone had a higher error rate (5.1% ±9.2%) in comparison to white skin tone (1.9% ±10.2%) for SpO_2 [56], [57]. Another study reported an increased error rate of 3.56 ± 2.45% among darker skin tone participants compared to white skin tone participants with an error rate of 0.37 ± 3.20% SpO_2 [58] , an issue that has more recently been researched [59], [60], [61]. Also, in a research study with anaemic participants during hypoxemia, it was observed that the error rate for the anaemic participants was 15% compared to 6.4% among non-anaemic participants [56], [62]. Surveyed literature disagrees on the effect of nail varnish, as some studies report a reduction of SpO_2 values [63], [64] (especially for black, blue, and green varnish [63]), whereas others found no effect [65], [66]. Next, we explore the novelty and process of rPPG working to help overcome PPG limitations.

C. Remote Photoplethysmography (rPPG)

The novelty in rPPG technology is its non-invasive nature to monitor physiological signs with no skin contact, as in Figure 2, addressing the limitations discussed in subsection I-B. As a non-contact technique, rPPG avoids the need for physical touch required by conventional PPG devices or smartwatches. This feature is advantageous for participants with cognitive or physical impairments, as discussed in subsection I-B. It is also a safer, non-invasive alternative to the ABG method for measuring SpO_2 levels (with risks listed in subsection I-A), and other vitals at a distance, with applications ranging from healthcare to fitness [47]. This approach measures changes in the blood flow and is based on subtle changes on the skin, where the pulse flashes lighter and darker over time. This phenomenon is not visible to the naked eye; however, it can be detected by measuring the reflectance of light over a period of time from skin pixel intensity extracted from Regions of Interest (ROIs) using a camera-based system [47]. rPPG systems measure the HR by analysing the skin pixel intensity of the heartbeats over time; the skin flashes darker and lighter as more and less blood flows through the region. SpO_2 is measured through the differences in optical absorption across the visible and near-infrared wavelength regions between Hb and deoxy-Hb [29], [67]. Various techniques from diverse research systems have been critically evaluated, summarised and presented in this survey.

II. Research Motivation

As seen, contact-based pulse oximeters pose challenges for many, which are mitigated with rPPG, a safer and more accessible solution to monitor multiple patients at a time. Despite its rapid growth, rPPG research still predominantly focuses on monitoring HR, with few studies simultaneously tracking both HR and SpO_2. To advance these systems their performance needs to be evaluated in various settings, including labs, clinics, and homes. To date, there is currently no comprehensive review conducted on the rPPG process for both HR and SpO_2 with an in-depth analysis of the limitations of rPPG systems, as reviews focus solely on HR measurement [68], [69], [70], [71], [72], with deep learning [73]. Most omit a comprehensive analysis of SpO_2 estimation methods [74], [75], [68], [70]. This review addresses these gaps.

III. rPPG PROCESS

Here we discuss the processes followed in rPPG systems. Surveyed articles used a variety of equipment and set-ups to obtain vital signs data from participants, such as thermal cameras [76], [77], [78], [79], Charge-Coupled Device (CCD) cameras [80], [81], other web cameras or those built-in laptops [82], [83], [84], [85], [86], [87], [88], [89], [90], iPads [91], MS Kinect V2 [92], [93], [67], Kinect Azure [94], GoPro cameras on drones [95], [96] and smartphones [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107] have been used to obtain a person's vital signs. Table IV and Table V [30] list studies conducted with various equipment and the vital signs under observation. The most common steps in rPPG systems are detailed below.

A. Face and ROI detection

This involves identifying the person's face from the frame, typically stationing participants in front of a camera in a static position to acquire face images. The participants are often required to be in a specific or fixed location with no or minimal movement. In the case of minimal movement, ROIs are selected manually [108], [109], [110] whereas to cater to head movement or rotation of the head to some degree, various algorithms are used, among which Viola and Jones [111], [112], [113], [114] is common. One main reason for using this library is its availability. Neural network-based classifiers have been used to detect ROIs [115], [116], [117], a statistical model to match a face to an image frame [118] and tracking algorithms of features points for head movement in image frames [119], [120]. One way of addressing the movement problem and identifying the correct ROI is to detect the face and ROI for each frame. However, the Viola-Jones bounding box is not very accurate and can report false positives and negatives [119] along with additional computation power in the case of real-time applications [119]. Kanade-Lucas Tomasi (KLT) was used by various researchers to track feature points from a face to update ROI location [115], [121], [122], [123], [124]. Research has also used Openface [125], [126] and Hyperbolic embedding graph structure [127], Multi-task Cascaded Convolutional Networks (MTCNN) [128] to obtain ROIs. Various parts of the body have been used as ROI to obtain vitals [129], [130], [131], [132], such as the entire face, forehead, and palm. Others have also segmented face into several patches to obtain better signals [133], [134], [67].

B. Signal extraction and preprocessing

Here, the raw signal from the ROI image data over a given period is extracted. Raw signals from a colour image include RGB channels and values obtained from each channel of the image frame are averaged over all pixels at a time period. Researchers have also used IR channel to obtain these values in addition to colour [135], [67]. For example, red channel pixels averaged for ROI image data for a time period of 60 seconds $(s_R(t), s_G(t), s_B(t))$ with s the signal obtained over time t for RGB channels. This process is called spatial pooling, and it helps average the noise contained in each pixel [47], [18], [136], [137], [138], [139].

C. Reducing noise, applying FFT and filtering

After the raw signal is obtained, the signal (S) containing data from all channels $[S = s_r, s_g, s_b]$ is passed through a Blind Source Separation (BSS), e.g., Independent Component Analysis (ICA) or Principal Component Analysis (PCA). This is applied as the signal can contain additional noise due to skin tone, light source or any movement. Applying BSS helps extract a clearer signal from the raw signal input [47], [18], [136], [137], [140]. FFT is applied to raw signals to convert them into the frequency domain. The signal is enhanced by removing unwanted frequencies, increasing Signal to Noise Ratio (SNR) [141], [142]. Bandpass filters are often used to remove low and high frequencies, e.g., 0.65 Hz and 3.5 Hz [47], [18], [136], [137], [143], [144], [145].

D. Extracting vitals and reliability checks

Vital sign data is extracted from the signal [47], [18], [136], [137], [146], [147], [148], [149], [150], [151]. A peak is identified from the signal to obtain HR data. HR is calculated by multiplying the frequency value on that peak by sixty to convert it into BPM, and for Respiratory Rate (RR), it is multiplied by the ratio of the number of peaks over a time period. Researchers used various methods to obtain SpO_2 from the pulse signal, which is mentioned in section V-B. Only two surveyed papers [67], [152] used SNR ratio and previous estimates within a window to perform consistency checks to ensure history-based reliability of vital signs data.

IV. METHODOLOGY

For this survey, we applied a meta-analytic approach to systematically evaluate the performance of non-contact HR and SpO_2 measurement systems on reviewed literature. We used PubMed, MedlinePlus, ACM, IEEE Xplore and Springer databases with the keywords: *remote*, *non-contact* and *heart rate monitoring*, *blood oxygenation level*, *rPPG*. The date range for the review was January 1990 – October 2022, from which 16,389 records were identified. This was refined to 2,416 after screening and then reduced to 187 (after removing 394 duplicate records, and applying the criteria in Table I). A second round was made based using the same criteria, to include research published in October 2022 – September 2023. In this round, 57 papers were added. A final round of review, added 132 publications from September 2023 – February 2024.

V. LITERATURE REVIEW FINDINGS: PREVIOUS WORK

This section provides details of surveyed research on rPPG systems, performance, conditions, and limitations. Most of the surveyed publications focus only on HR [153], a few focus on SpO_2 and much fewer focus on both. The system performance metrics (Root Mean Square Error (RMSE), r-correlation and Standard Deviation Error (σ)) of previous work are listed in Table III for HR and Table II for SpO_2 [30]. These non-contact-based rPPG systems have been validated by comparing their performance against clinical or commercial counterparts (i.e., contact-based ground truth devices). Research conditions and participant information are in Table IV and Table V (in Supplementary materials) [30].

TABLE I
INCLUSION AND EXCLUSION CRITERIA FOR rPPG LITERATURE REVIEW

Inclusion Criteria	Exclusion Criteria
Focus: On HR, SpO_2 and methods or techniques for measuring these signs.	On blood pressure and general health and activity monitoring. On security, perception, management, face detection and/or advantages of remote health monitoring. On economic, post-operative health monitoring involving carers.
Remote sensing with **camera-based** tech.	Contact-based monitoring, e.g., wearable (including pulse oximetry), wireless sensors or radars.
Protocol: Human participants able to consent, in a conscious state within a lab, clinic or home setting.	Experiments on animals or people who cannot consent for themselves. Measuring driver state or accident monitoring. Not reporting a complete experiment protocol, a complete system evaluation, or not peer reviewed.
Language: English.	Published in other languages.

A. Remote HR Measurement

We delve here into surveyed research measuring HR with rPPG, and their approaches for face identification, ROIs, signal extraction, participant movement, and other environmental characteristics, all under the lens of their effectiveness and potential for deployment in real environments.

1) Face identification: Thermal cameras can be used for rPPG data capture [76], [77]. Information about cardiac pulse, blood flow, and breathing rate can be extracted and the face movement tracked, with blood vessel registration used as ROI. After extracting the thermal signal and removing noise, a Fourier Transform is applied to obtain a pulse signal. The performance of one of such systems [76] is 88%, but RMSE, r-correlation or Mean Absolute Error (MAE) were not reported. Additionally, the data gathering took place in a controlled illumination and participants' movement was restricted (still state). The system might not perform well in a real-life scenario as illumination may vary and participants are not restricted from movement. Further, this thermal equipment can be prohibitively expensive [78], [119]. Similarly, Gatto et al. [78] used a thermal camera, with the forehead as ROI, to measure and analyse signal data to generate HR. During data acquisition, if the participant changed the head position in any direction, the system failed to capture data from the defined ROI. Researchers used the Auto-Regressive Moving Average (ARMA) model to obtain HR from the signal [154]. The main drawback is that the participant's movement in real-life can fail the system since it only tracks the forehead. In the case of a headscarf or longer hair, it is also highly likely to produce a reading that is not accurate. Along with that, a person's Body Temperature (BT) and environmental temperature could impact the signal extracted from a thermal video. The expensive thermal equipment would also make it unfeasible to deploy on a low budget [119], [155].

Remote HR measurement has also been achieved with CCD cameras. Researchers used 30 mean seconds of image data captured from the camera and analysed the change in brightness on participants' face regions from the cheeks. After analysing the signal data obtained from participants, HR peak was identified [80]. CCD camera and dual-wavelength LED lighting have also been used to measure HR but required a dark room and participants to have closed eyes, which may be impractical and uncomfortable [156]. Additionally, it's more vulnerable to ambient light and motion artefacts. Another study used a CCD camera to measure HR remotely. It focused on participants' movement and noise from light using chrominance rPPG to remove specular reflection variation. They used a combination of red and green channels to extract signals from defined ROI [81]. A study introduced a dual-bridging network to align domains and synthesise target noise, coupled with adversarial noise generation and hard noise pattern mining, aiming to boost rPPG model robustness [157]. Once again, these studies were conducted in a controlled setting or used public datasets with specific lighting. Deploying this system in a real-life scenario could impact the results (inaccurate readings) due to various types of illumination present in realistic environments. CCD camera used in this research is also expensive and bulky [119], [80].

Studies obtained HR used a laptop camera to acquire face colour images consisting of RGB channels. ICA, FFT and a frequency filter were applied to the channel data (RGB) to obtain the pulse signal. The second component (green channel) was found to typically produce the strongest pulse signal in this research study [82], [158], [159], [160], [161]. This was further refined through signal preprocessing such as detrending and smoothing the signal data. The system was updated to select the highest peak after applying FFT and filter, which was further smoothed by moving average filter [83]. However, the study was conducted within a lab setting (in a controlled environment) with a limited number of participants (n=12). In addition to that, restricted movement within a controlled environment does not replicate realistic environments. The face tracker used in the study detected multiple or no faces which the system handled by using the previous face captured; however, this method can fail in a real-life setting as it may not capture the real vital reading over time (using the previous face for a certain time) resulting in inaccurate HR readings.

2) Regions of Interest (ROIs): Several researchers used various ROIs from face images obtained from participants to measure HR remotely. Face ROIs included full face, forehead, lips, eyes or nose and cheeks, which varied in pixel size. These ROIs may contain regions which cannot provide information related to HR for example eyes, teeth, beards etc., which can impact the overall performance of results. In the case, where only the mouth area is focused to obtain HR, it can impact the error rate when a person is talking or laughing as teeth do not provide a pulse signal to obtain HR. To address this problem, researchers [115] introduced neural networks for skin classification [162] to determine skin colour from the face image. This ROI (skin pixel) was labelled and tracked using a mean-shift tracker, which iteratively shifts points of data to the average of points of data nearby [124]. The HR was obtained after applying FFT and used a data adjustment scheme which resulted in obtaining HR in a short time. This work was further extended using multiple video data, which reduced the error rate [163], though with no variety in light sources nor freedom of movement of participants expected in realistic settings.

3) Signal extraction: Researchers most commonly used PCA [164], [110], [165], [166], [167], [168], ICA [169], [115], [170], [171], [172], [173], [174], [175], [176], [177], [178], FastICA [53], [179], [180], [181], [182], [121], [183], [115], [170], [172], [173], [184], [185], [186], RobustICA [187], Joint Approximation Diagonalization of Eigenmatrices (JADE) [188] and also Laplacian Eigenmap (LE) algorithms [189] for extracting source signals from RGB channels to obtain HR. Wei et al. [189] used LE to extract signal sources from the face data obtained from participants. LE is also a dimensionality reduction technique but is non-linear to find the internal structure of the data. The research revealed better results in comparison to other dimensionality reduction techniques. Where one study suggested bypassing PCA or ICA to enhance processing speed, though this approach might not sufficiently reduce signal noise, potentially compromising effectiveness in real-world scenarios [190].

Eulerian Video Magnification (EVM) has been used to measure changes in the face region of the participants. The data was obtained in the form of a video that was spatially decomposed into various frequency bands to which temporal filtering was applied and finally amplified by a factor. This was implemented on data where participants were in a still state [191]. Other studies followed a similar approach to estimate HR and RR from video data as well [192], [193], [194], [195], [196], [197]. This method is not feasible for HR measurements in realistic settings, where a person could be in motion. In addition to that, this method may not work well when there is high noise present in the signal. Ciftci et al. [198] presented a method to generate facial videos with varied HR by synthesising effects on skin colour, aiming to enrich HR estimation research with sufficient annotated videos. However, ensuring the realistic synthesis of HR effects across all facial regions and skin types in synthetic videos remains a challenge [198]. Kossack et al. [199] evaluated tissue perfusion using locally resolved rPPG signals from RGB data to enable the detection of whether a tissue is adequately supplied with blood across different scales. However, challenges include analysing rPPG signals under changing lighting conditions, occlusions, and movements, affecting accuracy [199]. Other studies have also proposed methods for measuring [156], [200] and mapping rhythmical phenomena in tissue perfusion [201]. Some studies also used dual-wavelength measuring HR [202], [203], [204]. Research conducted also used machine learning techniques such as deep learning, Convolutional Neural Network (CNN), unsupervised and more to measure remote HR [205], [206], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [218], [219], [220], [221], [222], [223], [224], [225], [226], [227], [228], [229], [213], [220], [217], [218]. However, these studies need to ensure the real-world applicability of these models so it can be generalised across diverse environmental conditions and demographics for effective training. This can be a considerable challenge to extending these technologies to broad, practical applications.

4) Participant movement: Monkaresi et al. [230] allowed more movement among participants, but they used a machine learning approach to select the pulse component among all channels (RGB components) instead of choosing a single pulse component to identify peaks [82]. This approach includes applying power spectrum analysis, k-Nearest Neighbour (KNN) and linear regression to each component extracted after applying ICA to obtain features and classify them. KNN outperformed linear regression for selecting pulse signal components to obtain HR. However, generalising the model among the participants can bear different results or not perform well when new participants are introduced to the system on which it has not been trained yet and may therefore fail in a real-life scenario. Furthermore, the study was conducted in a controlled environment with specific light conditions.

To cater to challenges associated with different illumination and participant movement, Li et al. [181] used a technique based on Normalised Least Mean Square (NLMS) adaptive filtering and face tracking using the Viola-Jones face detector [231] to overcome these challenges. NLMS is an extension of the Least Mean Square (LMS) adaptive filter, which is used to simulate the selected filter. NLMS was used on raw signals to improve the data impacted by different light variations. To identify and locate face landmarks Discriminative Response Map Fitting (DRMF) the method was used in the initial image frame. KLT [232], [233] algorithm was then used to continue tracking the ROI. Researchers used the MAHNOB-HCI [234], [235], [236], [237], [238], [239], [240], [241], [242], [243], [244], [245], [246], [247] dataset for research studies to test their techniques. The results from their research exceeded the accuracy of previous studies. However, the database contains data obtained in a controlled environment with only slight movement in a lab setting. ROI tracking also fails when participants move at an extensive angle and participants' facial expressions create noise which then shows high variations in the obtained signal. Researchers have used datasets collected from lab setups to test and validate their methods, such as ARPOS [248], PURE [249], [250], [251], [252], [253], [243], [254], [255], [256], [257], [246], [247], [258], COHFACE [259], [257], [246], [247], MMSE-HR [260], [255], [246], [258], BH-rPPG [261], IIPHCI [262], [257], MPSC-rPPG [263], DOHA [264], OBF [265], [254], [245], [258], VIPL-HR [266], [267], [268], [244], [269], DEAP [270], [271] and UBFC-RPPG [272], [250], [251], [252], [253], [243], [254], [255], [246], [247], [258].

Bousefsaf et al. [273] used a method to select the most suitable ROI pixels, segmenting the face region into various sub-regions on the basis of lightness dispersal. The most suitable sub-regions were then chosen and grouped by assessing their SNR [273]. Another study focused on improving signal quality by reducing noise obtained from participants' face data by applying the block-based spatial-temporal on the face data. The spatial-temporal quality dissemination of the face ROI was then calculated depending on SNR. Adaptive ROIs were obtained by computing mean shift clustering and adaptive thresholding SNR maps, which then increased the accuracy of estimating HR [274], [275]. However, the system validation is limited to only controlled lab-based environments and needs to be stress tested for realistic environments. Laurie et al. [276] introduced an exposure control algorithm for optimising the SNR by adjusting the camera's exposure time to maximise it without distortion. However, the algorithm is sensitive to subject movement and has a longer execution time compared to standard controls. It requires participants to be relatively still to maintain accuracy, as movement can lead to colour channel saturation. Its longer execution time may limit its use in real-time or emergency situations [276]. A study used Kinect V2 and tracked skeleton to obtain HR and RR by monitoring the chest region movement with and without a blanket over a participant. This was to monitor cardiopulmonary irregularities such as bradycardia to process unclear ROI under well-lit and dark light conditions with different sleeping positions. In case abnormalities were detected, an alarm would be triggered to notify the carer [92]. However, the system may fail if a participant is sleeping on their stomach. Another constraint is the reliability of movement features to obtain the vital, which can create inaccurate observations for different types of motions such as walking, talking or micro facial gestures. These voluntary movements can impact the reliability of the vital data obtained. A robot with a video camera incorporated has also been used to obtain HR. The aim was to design a robot for older adults to do some exercise and measure their vital [249]; however, the accuracy drops with movement. Another research used a robot with colour and IR to measure HR [277]. However, these systems are not inexpensive which can be difficult to deploy on budget, especially for those within the middle to lower-income households around the world. Others capture rPPG from fingers [278], [279] which may be impractical when deployed in a real-life setting.

Most studies restricted participants from any movement [280], such as a person moving their head or walking around in the room, and even the motion of facial features such as while talking. In a real-life situation, these movements would be expected, which can increase noise when calculating vital signs. Research focusing on the head motion of participants has been addressed by researchers [109], [281], [282], [121], [283], [284], [285], [286], [287], [288], [289], [290], [291], [292], [293], [294], [295], [296], [297], [298], [299], [300], [301], [302], [303], [304], [305], [306], [307]. Where one of the researchers suggested tracking velocities of feature points on the face using Viola-Jones face detector and then tracking ROI by using KLT [109]. While other researchers focused on a single ROI, specifically the forehead [308], [309], [310], this was as micro expressions below the eye line are prone to more motion compared to the forehead [309]. Researchers have also addressed this movement-based HR monitoring by using Face Quality Assessment (FQA) to discard low-quality image frames, limiting the noise in the final obtained values. FQA uses four parameters, including resolution, luminance, sharpness, and face trajectory, to determine if the image requires to be removed. To track face feature points, the Good Feature Tracking (GFT) method was used on the image obtained. Landmarks from the face were obtained using the Supervised Descent Method (SDM) combined with GFT to obtain face feature points and track movement and hence provided improved signal quality to obtain HR [311]. However, these motions were restricted and conducted in a controlled manner within a lab setting. The systems deployed by the researchers had specific hardware requirements (e.g., large Random Access Memory (RAM) and Graphics Processing Unit (GPU) were needed) for these techniques, including controlled guidance to participants which are not feasible in real-life environments. Further, people from lower to middle-income backgrounds cannot take advantage of the system requiring expensive hardware [312], [313].

5) Other environmental concerns: In addition to the above, a single ROI such as the forehead can fail where a participant is not fully facing the camera. Further, various research methods focusing on measuring HR need to be validated within realistic environments to check how different factors such as illumination, facial hair (such as beard), makeup and frames per second (FPS) impact signal data used to obtain vital sign data [314], [315], [316], [317]. To increase SNR, some researchers use a monochromatic camera with a green range filter. This study also used weighted averages on the various face ROI. A deformable face fitting algorithm and KLT tracking [232], [318] algorithm tracked and extracted face features [50]. However, the RMSE obtained from this research is very high as shown in Table III, making it an inadequate method for real-life systems. Another study, by Zheng et al. [319], used EVM, Quality Assessment (QA) of signal, and Adaptive Chirp Model Decomposition (ACMD) to obtain HR. They validated the system in different illuminations and with some head motion. Researchers anonymised facial-rPPG videos to protect identity during vital sign measurement [320], [321], [322], [323]. However, its applicability may vary in different real-world conditions due to factors like lighting, movement, and camera quality.

Hasan et al. [324] introduced RhythmEdge which focused on model compression for use on devices with limited resources to obtain HR. It applies a multi-task learning framework to handle sensor differences, enhancing versatility. However, it faces challenges with camera variability, lighting, subject movement, and device limitations, which may affect measurement accuracy and consistency [324]. Different researchers also looked into ambient, natural and varying illumination [325], [326], [327]; however, once again, the setup was within a controlled setting where participants were at a distance of 0.6m from the camera. The participant diversity information was not revealed in the research paper. In addition to that, only the forehead is selected as ROI which would fail if the participant is not facing the camera.

Wang et al. [328] proposed a new continuous-spectrum IR illuminator using phosphor LED technology, addressing limitations of previous light sources. It offers improved colour homogeneity and energy efficiency for multi-wavelength Near Infrared Photoplethysmography (NIR-PPG) setups, useful in dark environments and non-obtrusive monitoring applications [328]. This research acknowledges limitations in the prototyping phase, including difficulties in achieving consistent colour homogeneity and adapting the illuminator to various spatial constraints and power needs. While effective in dark settings and with body motion, further validation is needed across diverse real-world conditions [328].

To address environmental challenges, particularly those posed by uneven lighting and colour variations, Song et al. [329] employed a combination of Ensemble Empirical Mode Decomposition (EEMD) and Multiset Canonical Correlation Analysis (MCCA) to specifically address spatially uneven illuminations. Decomposing the green signal and extracting the pulse enhances the robustness against uneven lighting. Ahmadi et al. [330] used amplitude-selective filtering (ASF) and colour-distortion filtering (CDF) to counteract the effects of spatial variations in skin colour and lighting. Whereas Lu et al. [331] introduced NEural STructure modelling (NEST), aimed at enhancing the generalisation of rPPG models across different unseen environments. NEST tackles the domain shift challenge in rPPG tasks through three main strategies: optimising feature space coverage, aligning multi-domain features at an instance level, and enriching discriminative features. Only one study [67] was found that was conducted in participants' homes, involving different camera-to-subject distances, and shared its open dataset. However, it reported a higher error rate, particularly among individuals with darker skin tone and those in an active state [67]. van Esch et al. [332] used RGB cameras in a hospital environment, applying machine learning to detect cardiac arrhythmia, including atrial fibrillation and flutter, without limiting patient movement. Although it is a significant advancement in non-invasive cardiac monitoring, the study highlights a persistent challenge: the difficulty in accurately differentiating arrhythmia due to their similar signal characteristics, especially when patients move. This issue underscores the need for refinement of the detection models.

B. Remote SpO_2 Measurement

Relatively few studies used methods to obtain SpO_2. One method, cancelling aliased frequency components induced by light flickering (based on autoregressive modelling and pole cancellation), improved its effectiveness under fluorescent illumination [333]. The research was conducted on patients undergoing kidney dialysis (at rest, with minimal movement) [333]. However, movement and illumination change increased noise, in turn impacting system accuracy. The research also does not provide RMSE, r-correlation or σ for SpO_2. While using a webcam, another research measured SpO_2 and HR by using an algorithm for noise removal based on Dual-Tree Complex Wavelet Transform (DTCWT) to fix motion artefacts and artificial illumination [33]. RGB channels have been used to obtain SpO_2 by assessing the pulse signal at two wavelengths of 660 nm and 940 nm [334], comparing red and blue wavelength bands and using a 50 cm range [33], [334], [335].

A study used wavelet transform denoising to address lighting and movement artefacts. It was validated for healthy individuals and Pediatric Intensive Care Unit (PICU) patients, noting the impact of Hb levels on estimation accuracy. Despite showing strong correlations with oximeter readings, the method faced challenges in accurately estimating values for anaemic patients and requires further validation across varied clinical conditions[336]. Kim et al. [337] measured SpO_2 using RGB camera and YCgCr colour space, enabling estimation in ambient light without direct contact. It leveraged absorption differences of Hb for SpO_2 analysis. However, movement, lighting, and makeup on signal quality, and its validation is limited to SpO_2 levels of 85–100%, indicating a need for broader testing [337]. Another research used RGB webcams, focusing on enhancing noise resistance and accuracy through BSS and energy coefficient analysis. However, it has only been validated on a small and homogeneous group of participants. Additionally, as with most research, this needs to be validated in a real-life setting to further evaluate the system [338]. Stogiannopoulos et al. [339] used IR and EVM to highlight skin pixel intensity variations for nighttime monitoring to obtain SpO_2. It evaluates machine learning regression techniques like the Generalised Additive Model (GAM) and Extra Trees Regressor, alongside deep learning approaches including 3D CNN and Video Vision Transformer (ViViT), on amplified PPG signals from facial videos demonstrating an acceptable range of accuracy. However, validation primarily under controlled conditions and the method's adaptability to varying equipment and environmental settings may be a challenge when replicating to real world scenarios [339].

Researchers also measured SpO_2 using ambient light and two cameras equipped with narrow-band filters at wavelengths of 660 nm and 520 nm. The need for precise alignment and adjustment due to the use of two separate cameras and potential challenges in filtering out motion artefacts and other interferences in ambient light conditions is a challenge. The method also depends on the ambient light's intensity and spectrum, which might limit its applicability in varying environmental conditions [340]. Verkruysse et al. [341] used digital RGB cameras and Monte Carlo simulation for light transport analysis. This was validated in controlled conditions for an SpO_2 range of 83% to 100%. The study acknowledges limitations, including its validation in a controlled environment, a small sample size, and challenges in adapting to uncontrolled conditions like variable lighting and movement [341].

Another approach [67] to measuring SpO_2 uses the extinction spectra of oxy-haemoglobin and deoxy-haemoglobin, via a ratiometric measurement technique, dividing the recorded pixel intensity between two distinct spectral ranges, red (600–700 nm) and IR (800–900 nm) to obtain this vital data. The reported error rate was consistent over all participants of diverse skin tones in resting and active states of ±2%, but this was not yet clinically validated. To obtain SpO_2, Al-Naji et al. [342] used Ensemble Empirical Mode Decomposition (EEMD) and ICA. However, the system's use of visible light cameras may face issues like unclear ROI from movement and skin tone challenges. Additionally, further real-world validation is needed to confirm its practicality [342]. Other researchers used a Complementary Metal-Oxide-Semiconductor (CMOS) camera and dual-wavelength LED illumination at optimized wavelengths (611 nm and 880 nm), showing strong accuracy and consistency across a clinically relevant SpO_2 range. Despite its innovation, limitations include hardware dependency, sensitivity to environmental conditions, and a validation range of 83%-100% SpO_2, suggesting a need for further clinical and real-world validation [343]. Xi et al. [344]

employed singular spectrum analysis (SSA) for signal decomposition and a spectral masking algorithm to measure SpO_2. Once again, it has been tested in controlled settings and requires validation for motion and skin tone [344]. Researchers also used a deep learning approach for measuring SpO_2 by employing a spatial-temporal map (STMap) for data encoding. However, challenges such as the need for broader clinical trials and data standardisation were highlighted [269]. One approach uses hand palms to measure SpO_2 by applying spatial averaging, obtaining RGB time series and applying a CNN [97], [345]. However, using palms or fingers under a camera is impractical in real-life due to the need to keep hands still. Additionally, hands are a less exposed part of the body compared to the face, which would make it difficult to deploy it in real-life scenarios [97], [345], [278].

A variety of equipment has been used for rPPG, including CCD cameras, smartphones, laptops' webcams, drones with cameras and Kinect. Each device has its characteristics such as resolution, dimensions, processing power, data type collected and cost etc. Low and high-resolution digital cameras, and smartphone cameras capture RGB channels have been used in most studies to obtain vital values; whereas thermal cameras allow capturing the thermal data of a participant. However, smartphones need to be held continuously, which can be impractical for long periods. Also, the cost of thermal [119], [78] and CCD equipment [80] may be prohibitive for certain deployments (as discussed in section V-A).

VI. Limitations of Existing rPPG Systems

In what follows, limitations of reviewed rPPG systems are summarised, with areas for further research and improvement.

A. Equipment limitations

Thermal and CCD cameras used in these studies can be prohibitively expensive [119], [78], [312], [313], [155], [346]. A problem with using a thermal camera is that it can generate temperature from the body and also the environment, which can impact the data acquired from a participant, such as thermal noise [119]. Furthermore, ambient sensing of oxygen saturation through multispectral imaging has been explored but using specialist scientific equipment which can be expensive to measure SpO_2 [347], [348], [349], [350], [351]. These studies have primarily been conducted in laboratory settings, utilising specialised hardware with ample RAM and GPU resources. However, this lab-based approach may not readily translate to real-world environments. In clinical or home settings, individuals may be positioned at varying distances from the equipment, making it essential for the system to deliver accurate readings under diverse circumstances. All of the research studies (except one [67] conducted among participants' homes) used specific hardware to acquire data with camera FPS set at 15, 30 or 60 as shown in Table V [30]. This ensured a constant FPS for each RGB channel. The system (code) and equipment (computer and camera) were set up by the researcher in all studies and had enough processing power, RAM, graphics card and storage space to collect this data. However, people in a real-life scenario may not have access to expensive scientific equipment where the RAM, graphics card and GPU may impact the FPS and in turn, impact the number of samples in a signal obtained for each channel and if the system does not cater to this, it may impact error rate of the system. FPS impacts signal quality that is used to obtain vital data obtained data from realistic scenarios [67]. Bit depth, with 8 bits per channel (multi-channel), is prevalent; however, a higher bit depth could yield a cleaner signal. Developed rPPG systems report using various resolutions and equipment, as shown in Tables IV and V (in supplementary). Quantum efficiency needed for this improvement remains uncertain and needs further research. Additionally, the precise impact of parameters such as resolution and bit depth on accuracy is not well understood, indicating further research.

B. Controlled Environments

rPPG studies have predominantly been conducted within controlled environments where only one study evaluated their system in participants' homes. While it is important to initially test and validate these systems in controlled environments, it is equally crucial to evaluate their performance in real-world scenarios, particularly within clinical settings. Clinical environments encompass a wide range of HR and respiratory conditions, involving diverse participants. Real-world validation against clinical standards is vital for ensuring the effectiveness and reliability of such systems. Researchers need to ensure they share evaluation measures when discussing the results of their research systems to progress in the field.

Currently, there is no commercially available (and clinically validated) rPPG systems able to monitor these vital signs. It is crucial for such systems to be validated in real-life environments including labs, homes and clinics to stress these systems. Furthermore, the effectiveness of these systems for monitoring health in clinical settings has yet to be fully validated. It is crucial that rPPG systems demonstrate the ability to promptly identify patients at the highest risk of health deterioration, highlighting the significant advantages of utilising such remote technology in healthcare.

C. ROI Selection and Occlusion

Using a single Region of Interest (ROI) can prove ineffective in measuring vital signs. This limitation becomes evident when the chosen ROI is obscured from the camera's view, for instance, when a person wears a headscarf or has long hair that conceals the forehead, or when the individual is positioned at an oblique angle to the camera. Therefore, it is crucial to test systems comprehensively across diverse settings to validate their performance and evaluate their error rate.

Tracking people can also be occluded by objects, other people or both from the camera's view to send data to the system which is common for front-viewing cameras used by the systems [352]. Due to occlusion, a user's skeleton cannot be identified; however, there are several methods to address this issue, including hand over face occlusion [353], API tracking of multiple people's positions in environments where occlusion occurs [354], [355], [352], location-aware wearable haptics [356], sound localisation [357], using API, a toolkit to track and multiple cameras within an environment to track multiple peoples and joints using depth data [352] and using shadow and skeletal fusion data to track people within an environment [358]. Occlusion in rPPG systems needs to be further studied.

D. Participant Diversity and Experiment Conditions

Most of the surveyed research in Table IV and Table V from [30] have a small number of participants and a small variation in skin tone. For SpO_2 specifically, all studies included fewer than 15 participants (except one [67] as shown in Table II from [30]). Similarly, for HR measurement, most studies had a small number of participants [359], [360], [361], [362], [363], [364], [365]. Only three studies stood out with a larger number of participants [319], [67], [81]. Among these, two studies involved 40 participants each [319], [67]. However, only one of these studies collected vital sign data within realistic home environments, encompassing both HR and SpO_2 measurements [67]. The third study [81], which included 117 participants, focused on measuring HR only and was conducted in a controlled environment resembling a studio setup with limited movement and using expensive equipment [81]. The remaining studies surveyed in Table IV had much fewer participants and strict experiment conditions. Furthermore, most of the surveyed studies in Table V did not measure oxygenation levels at a distance [30].

The surveyed literature considered only one participant at a time (with exceptions, with up to six participants [47], [67]). Measuring multiple participants simultaneously is required, especially in shared environments. All the research studies mentioned above had participants closer to the camera (up to 0.5 m from the camera). Very few studies measured HR up to 4 m [92], [67], up to 60 m [366] and up to 100 m [367]. For participants whose HR was measured at up to 100 m, light reflectance was lower. Reduced ROI at long distances also impacted accuracy as shown by large mean absolute errors with greater imager-to-subject distances [367]. Most studies do not disclose participant characteristics and typically focus on a specific demographic [330] limiting the generalisability of the findings, including women, older adults, or individuals with cardiovascular conditions. A wider range of demographics and skin tones need to be included in research studies.

Realistic environments are varied (e.g., homes and triage rooms in clinics and hospitals) and various factors such as facial expressions, rotation, distance from the camera, illumination, facial hair and skin tone can impact the system's obtained data. Parameters related to makeup and beard were only reported in two publications, albeit with a small number of participants [368], [67] and one study also does not analyse or segment results based on these parameters [368]. These experiments ought to be conducted with diversity in mind (both environmental and experimental).

E. Datasets

Most of the surveyed studies did not share data or code. Publicly available datasets were collected under lab conditions with restricted parameters such as light, movement, and fixed camera distances. There is a lack of publicly available data collected from a real-life environment, so the system evaluation measures can be tested on those to evaluate their system [47], [67]. Only one dataset includes data from home environments, yet this did not include participants with black skin tone [67]. Studies conducted restrict participants from micro facial gestures or physical movement during vital data acquisition. Additionally, if the error rate is not evaluated for realistic environments with natural movement or for example during undergoing treatment, it can result in failure to accurately measure vitals due to the presence of noise. Therefore, it is essential to measure vital sign data in real-life environments [47], [67]. Face tracking will also impact ROI selection when movement takes place, depending upon the lighting, and exposure length of the camera, it could potentially create blurry frames, and it is important to analyse data with noise to ensure the system will be able to perform as required in a real-life environment [47]. It would be more beneficial if video streams of Colour, IR and Depth Videos would be made available open source so researchers can analyse the realistic data including various movements, illumination, environments, skin tone and other factors with their respective timestamps to analyse their algorithms.

VII. Bias in rPPG Systems

The use of camera-based technology to capture facial data poses some challenges in rPPG systems, in particular, due to the diversity of skin tone in humans [369]. rPPG systems must be highly accurate for clinical use. Therefore, rigorous validation and extensive testing across a diverse range of participants are imperative. Many publicly available databases primarily emphasise individuals with lighter skin tone. Neglecting diversity can pose a significant disadvantage when deploying rPPG systems; making it beneficial only for a specific group of people. One study [50] reported a higher error rate in such cases. Another study [67], reported a lower error rate and introduced effective techniques across a range of skin tones. A critical observation from this review is the limited consideration given to participants who wear makeup (e.g., lipstick or concealer), or those with facial hair. These factors can significantly influence the accuracy of data readings in rPPG systems. Only two research studies were found that stated participants wearing makeup [67], [370]. Both studies found makeup in colour and IR impacted the accuracy of the rPPG systems and suggested further rigorous testing. However, only two studies [67], [370] reported participants wearing lipstick or foundation. Consequently, it becomes important to rigorously validate these systems, accounting for bias, and further evaluating their performance across a wide demographic.

VIII. Summary

This paper offers an in-depth review of rPPG-related research that can measure HR and/or SpO_2. Processes, techniques, participant characteristics, equipment and experiment conditions used in the surveyed research were discussed, as well as the performance, evaluation measures and limitations of these studies. The majority of the studies focused on measuring HR, a few on SpO_2 and very few on both. Previous research needs to consider various factors when designing an rPPG system including varying camera-to-subject distances, fluctuating frames per second, computational resource requirements, diverse environments, illumination conditions and participant diversity. Further, it is crucial to validate systems under conditions involving natural movements during vital sign measurements to improve accuracy. This issue is critical when considering real-world deployment and clinical use.

In conclusion, the field of rPPG has made significant progress in recent years. We have highlighted various aspects of these methods for measuring HR and SpO_2, identifying several important findings and research challenges. Addressing these challenges is crucial for advancing this field, as rPPG has great potential for clinical and non-clinical applications. This review serves as a valuable resource for researchers seeking to further improve the accuracy and reliability of these systems in real-world scenarios as this continuously evolving field grows.

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